Energizing Power Superhighways in India

On its green journey, India faces the pressing challenge of ushering in intelligent transmission systems. HVDC is a key enabler of carbon-neutral energy systems as it is capable of efficiently transmitting a large quantum of electricity over long distances with minimal footprint while ensuring the integration of renewables through interconnecting grids. Hitachi Energy is well poised to address this challenge with its unparalleled track record in HVDC projects, including the pioneering Vindhyachal project, which marked the company's introduction of HVDC technology in India. Achieving a breakthrough in pioneering bi-directional transmission links, Hitachi Energy commissioned the +/-800kV Raigarh-Pugalur Ultra High Voltage Direct Current (UHVDC) link. Spanning 1,800 kilometers and capable of serving over 80 million people, this link stands as a model of sustainability, that can deliver a 180MW energy savings, reducing CO2 emissions by approximately 646,500 tons per year, and preserving around 130 square kilometers of prime and forested land. Pushing the boundaries of power transmission capabilities, Hitachi Energy is also setting up an HVDC infeed link in Mumbai to supply up to 1,000MW of electricity, which can increase power from outside of the city by almost 50 percent. The city centre infeed link, built with a combination of underground cables and overhead lines, addresses the city's soaring electricity demands.

HITACHI
Inspire the Next

Our heritage: 75 years of powering India

75 years of Powering India

1949 Hindustan Electric Company is incorporated

1960-61 Expands manufacturing of transformers and HT air blast circuit breakers <220kV in Howrah and Baroda (vadodara) as part of a technical collaboration with Brown Boveri & Cie (BBC)

1962 New manufacturing facility set up in Maneja, Baroda - the flagship **circuit breaker** factory

Hindustan Electric Company name changed to Hindustan Brown Boveri Ltd after merging with BBC



1984 ASEA started manufacturing in Bangalore (Bengaluru)

1988 ASEA and BBC merge globally

1989 Hindustan Brown Boveri Ltd name changed to Asea Brown Boveri (ABB) Ltd.

Commissioned first **back-to-back HVDC transmission link** for NTPC at Vindhyachal, connecting northern and western grids

1990 Commissioned India's **first long distance (814kms) HVDC link** Rihand - Delhi HVDC 1,500 MW, ±500kV

2007 India Operations Center for Power Grids Business launched in Chennai

2015 Installed end-to-end power solution for **Delhi Metro**, and **SCADA** to monitor and control the power network

2016 Energized one of the highest rated **1,200kV transformer**

2017 Commissioned the **world's first multi-terminal UHVDC link** - the 6,000MW, 80kV DC link connecting North-East India to Agra

2017 **First digital substation** for reliable, round the clock power to serve the 350 companies for India's largest Information Technology park

2018 Partnering Indian Railways' electrification ambitions with extensive traction equipment orders

2019 Booked first order for cyber-security from POWERGRID

ABB Power Products & Systems India Ltd. incorporated

2020 Power Grids business acquired by Hitachi from ABB

Listed on the National Stock Exchange and BSE

2021 Name change from ABB Power Products and Systems India Limited to "Hitachi Energy India Limited" thus Hitachi Energy is formed

2022-2023 Inaugurated HVDC & Power Quality factories in Chennai & Bengaluru; GTIC in 2023

Electrifying the Future: Enabling Clean Mobility in India

The rapid electrification of industries is reshaping the energy landscape, with mobility emerging as a crucial sector in this transition. Indian Railways, India's mobility lifeline, is on a fast track to achieve 100% electrification of its network, with a vision to become the world's largest green railway with zero carbon emissions by 2030.4 Hitachi Energy is at the forefront of this transformation, leveraging its highest-rated Scott transformer of 100MVA. Designed and manufactured locally at its Maneja facility in Vadodara, these high-capacity transformers play a critical role in upgrading the Indian Railways network, facilitating the transition from traditional electrification systems to more efficient and reliable solutions. By supporting higher-speed and more frequent train services, these transformers not only enhance mobility but also contribute to significant reductions in carbon emissions, aligning with Indian Railways' goal of achieving net-zero carbon emissions.

Meanwhile, India's push for 30% EV sales by 20305 highlights the need for robust charging infrastructure. The Grid-eMotion® Fleet solution can enable this by optimizing energy consumption and scaling EV operations, to support a cleaner, more sustainable transportation future.

Sources

1. https://t.ly/4RJ6a (IBEF)
2. https://t.ly/WHb-A (PIB)
3. https://t.ly/IzY36 (NewsOnProjects)
4. https://t.ly/dRdkO (Financial Express)
5. https://shorturl.at/femwv (The Economic Times)

"Hitachi Energy has grown with the nation, understanding its energy needs to develop industries and infrastructure through seminal nation-building projects which have amplified our focus on social innovation. Our legacy demonstrates our unwavering dedication towards bringing a clean, flexible, and sustainable energy future for all in the country. As India grows to take its place as the third-largest economy in the world, there will be enhanced focus on sustainable growth and the need to secure our net-zero ambitions. Advanced digital technologies will play a pivotal role in navigating the complexities of the evolving energy landscape while ensuring sustainability and creating a strong & resilient grid. Electricity will be the backbone of the entire energy system and with electrification being pivotal to achieving net-zero, our solutions, services and innovations will set the tone for the country's energy transition."

N Venu
MD & CEO, India & South Asia, Hitachi Energy



Scan for Social Innovation Business Site